

FLEA NEWS 28

Department of Entomology
Iowa State University
Ames, Iowa 50011 U.S.A.

JUNE 1984

Compiled by Robert E. and Joanne H. Lewis

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Please find herewith:

---Lists of additional literature for 1973 (list 14), 1974 (list 17), 1975 (list 17), 1976 (list 16), 1977 (list 14), 1978 (list 15), 1979 (list 14), 1980 (list 10), 1981 (list 8), 1982 (list 5), 1983 (list 3), 1984 (list 1).
---additions, changes and deletions to the mailing list.

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It is with deep regret that we announce the death of Dr. V. E. Tiflov. Dr. A. I. Goncharov has reported that Dr. Tiflov, born in 1899, died on March 30, 1984.

OBITUARY

C. Andresen Hubbard
10-XII-1897 30-IV-1984

It was with considerable sadness that we were informed by Mrs. C. A. Hubbard of the death of her husband.

Dr. Hubbard was born at Davenport, Iowa, and christened Clarence Albert. In 1904 his family moved to Portland, Oregon, where he graduated from the public school system in 1917. The same year he enrolled in the University of Washington, Seattle, receiving his Bachelor of Education degree in 1920 with a minor in Zoology. From 1920 to 1922 he was a teaching fellow at the University of Washington. In 1922 he assumed the position of Head of the Biology Department, Pacific University, Forest Grove, Oregon, functioning in this capacity until 1942. During this period he continued his graduate education and, in 1932, earned a double Masters degree in Education and Zoology from the University of Washington. In 1937 he was awarded the degree of Doctor of Science by Pacific University for his dissertation entitled "Fleas of Western North America", which formed the basis for his book of the same title published by the Iowa State College Press in 1947. He was a biologist with Vanport College (now known as Portland State University), Portland, Oregon, from 1943 to 1946 and in 1944 became Head of the Department of Biology at Warner Pacific College, Portland, occupying this position until 1962. During this period he received a Fulbright grant to study fleas and plague in Iraq (1953) and subsequently a National Science Foundation grant to study the fleas of northern California and East Africa. On another Fulbright grant in 1962 he went to the Malaria Institute at Amani, Tanga, Tanzania, where he initiated studies on

C. Andresen Hubbard

fleas and plague in that area. These studies were further supported from 1962 to 1967 by another grant from the National Science Foundation. In July 1967, he entered retirement at Tigard, a suburb of Portland, Oregon.

During his professional career Dr. Hubbard published 61 papers, 51 of which dealt with fleas or related subjects (see Flea News 26 for bibliography). He also described 76 new flea taxa on the species level, 15 of which are currently considered to be junior synonyms. He erected 3 genera, Augustonius, Corypsylloides and Foxelloides, all of which are currently considered to be synonyms.

In his later years Dr. Hubbard, or simply Hubbard, as he signed his letters, always insisted that fleas were secondary to him and that his main interests resided in their mammalian hosts. In spite of this, the bulk of his publications dealt with the flea fauna of the western United States, and his is the only comprehensive review of this fauna that has ever been published.

Hubbard was an excellent field mammalogist, returning to collecting localities often to add additional records of the rarer species. He was occasionally mildly criticized by some of his colleagues for his use of the vernacular in his writings, but they have stood the test of time and have yet to be replaced by superior works.

In the words of a colleague who knew him personally, "Hubbard was a great old gentleman and I will miss him and his interesting letters".

Vital statistics used here were provided by Dr. Hubbard in an autobiographical sketch dated 1-August-1980. A chronological list of the flea taxa he described and their current status follows.

---R.E.L.

- jordani Hubbard, 1940 Nearctopsylla [now N. (Beringiopsylla) jordani]
jordani Hubbard, 1940 Corypsylla
kohlsi Hubbard, 1940 Corypsylla
jellisoni Hubbard, 1940 Doratopsylla = D. curvata obtusata Wagner, 1929
jordani Hubbard, 1940 Epitedia = E. scapani (Wagner, 1936)
stewarti Hubbard, 1940 Epitedia
sculleni Hubbard, 1940 Catallagia
chamberlini Hubbard, 1940 Catallagia
motei Hubbard, 1940 Catallagia = C. charlottensis (Baker, 1898)
walkeri Hubbard, 1940 Meringis = M. hubbardi Kohls, 1938
jewetti Hubbard, 1940 Meringis = M. hubbardi Kohls, 1938
dobbsi Hubbard, 1940 Malaraeus [now Amalaraeus dobbsi]
jellisoni Hubbard, 1940 Thrassis [now Oropsylla (T.) pandorae jellisoni]
goodi Hubbard, 1941 Micropsylla [now Rhadinopsylla (M.) sectilis goodi]
rockwoodi Hubbard, 1942 Thrassis [now Oropsylla (T.) francisi rockwoodi]
jamesoni Hubbard, 1943 Meringis
kansensis Hubbard, 1943 Monopsyllus exilis = Pleochaetis exilis (Jordan, 1937)
princei Hubbard, 1943 Thrassis = T. fatus (Jordan, 1925)
intermedius Hubbard, 1943 Orchopeas sexdentatus
pacifica Hubbard, 1943 Dactylopsylla [now Spicata pacifica]
bottaceps Hubbard, 1943 Dactylopsylla [now Spicata bottaceps]
arizonensis Hubbard, 1947 Foxella utahensis [now F. (F.) ignota arizonensis]
kincaidi Hubbard, 1947 Monopsyllus ciliatus [now Ceratophyllus (Amonopsyllus) ciliatus kincaidi]
oregonensis Hubbard, 1947 Megararthroglossus procus = M. procus J. & R., 1915
wallowensis Hubbard, 1947 Megararthroglossus divisus = M. divisus (Baker, 1898)
clantoni Hubbard, 1949 Megabothris [now M. (Amegabothris) clantoni clantoni]
johnsoni Hubbard, 1949 Thrassis gladiolus [now Oropsylla (T.) bacchi johnsoni]

princei Hubbard, 1949 Megabothris [now M. (Amegabothris) clantoni princei]
johnsoni Hubbard, 1949 Megabothris clantoni [now M. (A.) clantoni johnsoni]
moorei Hubbard, 1949 Dactylopsylla [now Spicata moorei moorei]
clantoni Hubbard, 1949 Foxella ignota [now F. (F.) ignota clantoni]
traubi Hubbard, 1949 Nearctopsylla jordani [now N. (Beringiopsylla) traubi]
augustsoni Hubbard, 1949 Thrassis [now Oropsylla (T.) augustsoni]
hoffmani Hubbard, 1949 Thrassoides [now Oropsylla (Thrassis) aridis hoffmani]
kylei Hubbard, 1949 Monopsyllus wagneri [now Aetheca wagneri (Baker, 1904)]
charlestonensis Hubbard, 1950 Monopsyllus eumolpi = Eumolpianus cyrturus (Jord. 1929)
wallowensis Hubbard, 1950 Monopsyllus eumolpi [now Eumolpianus wallowensis]
canadensis Hubbard, 1950 Monopsyllus eumolpi = E. eumolpi (Rothschild, 1905)
americanus Hubbard, 1950 Monopsyllus eumolpi [now E. eumolpi americanus]
scapoossei Hubbard, 1954 Dactylopsylla comis [now Spicata comis scapoossei]
tacomae Hubbard, 1954 Dactylopsylla comis [now Spicata comis tacomae]
walker Hubbard, 1954 Dactylopsylla comis [now Spicata comis walker]
oregona Hubbard, 1954 Dactylopsylla moorei [now Spicata moorei oregona]
washingtona Hubbard, 1954 Amphipsylla sibirica
eaton Hubbard, 1954 Oropsylla arctomys [now O. (Oropsylla) eaton]
foust Hubbard, 1954 Thrassis = O. (T.) spenceri (Wagner, 1936)
martyoungi Hubbard, 1954 Nearctopsylla [now N. (Beringiopsylla) martyoungi]
fasten Hubbard, 1954 Monopsyllus ciliatus = M. c. kincaidi Hubbard, 1947
reeher Hubbard, 1954 Monopsyllus vison = C. (M.) vison (Baker, 1904)
cochisei Hubbard, 1954 Meringis
fireman Hubbard, 1955 Orchopeas sexdentatus
smiti Hubbard, 1956 Coptopsylla
allousei Hubbard, 1956 Ctenophthalmus congener [now Ctenophthalmus (Eu.) allousei]
johnsoni Hubbard, 1956 Chiropteropsylla brockmani
pringlei Hubbard, 1956 Nosopsyllus [now N. (Gerbillophilus) pringlei]
durii Hubbard, 1956 Nosopsyllus [now N. (Nosopsyllus) durii]
bunii Hubbard, 1956 Nosopsyllus [now N. (Gerbillophilus) bunii]
linni Hubbard, 1958 Kohlsia
grayi Hubbard, 1958 Jellisonia
chapmani Hubbard, 1958 Foxella ignota [now F. (F.) ignota chapmani]
morgandavisi Hubbard, 1963 Xenopsylla
raybouldi Hubbard, 1963 Xenopsylla
serengetiensis Hubbard, 1963 Xenopsylla sarodes
manyarensis Hubbard, 1963 Xenopsylla sarodes
wilsoni Hubbard, 1963 Libyastus
titan Hubbard, 1963 Dinopsyllus [now D. (Dinopsyllus) titan]
pringlei Hubbard, 1963 Dinopsyllus [now D. (Dinopsyllus) pringlei]
leptodactylus Hubbard, 1963 Ctenophthalmus [now C. (Ethioctenophthalmus) ...]
gilliesi Hubbard, 1963 Ctenophthalmus [now C. (Ethioctenophthalmus) ...]
hopkinsi Hubbard, 1963 Ctenophthalmus particularis [now C. (Idioct.) hopkinsi]
hemingwayi Hubbard, 1963 Ctenophthalmus cophurus [now C. (Ethioctenophthalmus) ...]
wilkesi Hubbard, 1963 Ctenophthalmus evidens [now C. (Ethioctenophthalmus) ...]
grzimeki Hubbard, 1964 Ctenophthalmus [now C. (Ethioct.) cophurus grzimeki]
schmiederi Hubbard, 1966 Ctenophthalmus cophurus [now C. (Ethioct.) ...]
smiti Hubbard, 1967 Ctenophthalmus particularis [now C. (Idioct.) hopkinsi smiti]
smithi Hubbard, 1967 Xiphiopsylla lippa

* * *

RECENTLY PUBLISHED - The Rothschild Collection of Fleas. The Ceratophyllidae: Key to the genera and host relationships. With notes on their evolution, zoogeography and medical importance by R. Traub, M. Rothschild and J. F. Haddow, 1983. (ISBN 0-12-697680-5) Cambridge University Press/Academic Press (Cambridge and London), i-xv, 288 pp. + 90 pls., 205 figs., 175 maps (£ 60.00, \$99.00)

It has generally been assumed that the last volumes of the monumental Catalogue of the Rothschild Collection of Fleas by G.H.E. Hopkins and M. Rothschild would treat the complex family Ceratophyllidae. It has also been assumed that at least two and, more likely, three volumes would be required to cover the entire family. Early works by Jordan (1933), Wagner (1934) and Ioff (1936) began the process of bringing order to the family, but these authors suffered from lack of material and knowledge of the world fauna as we know it today. For example, Ioff stated that he examined only 69 species. Since 1936 over 400 additional ceratophyllid species have been described. For these and other reasons, the authors of the volume reviewed here deemed it impractical to follow the Catalogue format established in the preceding six volumes and this is not to be considered as part of the Catalogue series. The result is an assemblage of articles by various authors, or combinations thereof, which seems likely to confound pulicologists and bibliographers alike for generations to come. Although of somewhat smaller dimensions than the Catalogue volumes, the title is very similar and the binding identical in both substance and color.

Beyond these minor criticisms, the book contains a wealth of information and certainly represents the best synthesis of information for any family of the order Siphonaptera. Due to the varied nature of the different sections, it seems appropriate to review them separately. After a brief Introduction describing the scope of the volume, data arrangement, explanation of maps and map captions, data limitations and notes on the higher classification of the order, are Acknowledgements and the following sections.

SMIT, F.G.A.M. 1983

Key to the genera and subgenera of Ceratophyllidae. pp. 1-36. 250 figs. 90 pls.

This section begins with a brief characterization of the family and a comparison of it with the Leptopsyllidae. Following this are notes on the morphology of members of the family, beginning with the cranium and concluding with the modified abdominal segments in both sexes. Then come descriptions of the 7 new genera and 8 new subgenera erected here, and a key to the genera and subgenera currently recognized as belonging to the family. After each genus are listed all of the valid species, with the generitype or subgeneritype indicated by an asterisk (*). The 250 figures are grouped at the end of the key. After these come 90 plates, each containing 5-6 superbly retouched photographs of a representative species of each genus and subgenus, arranged in alphabetic order. Those familiar with recent illustrated works by Traub are well aware of the effectiveness of such illustrations, and presumably these were contributed by Dr. Traub for use in this volume.

TRAUB, R., M. ROTHSCHILD & J. F. HADDOW 1983
Definition of zoogeographical areas. pp. 37-41

In this section the authors follow the classical arrangement of zoogeographical regions put forward by Bartholomew, Eagle Clarke and Grimshaw (1911) in their Atlas of Zoogeography, with the addition of a Wallacean Region for Celebes, Timor, Flores, Sumba, Halmahere, Ceram and Buru. Only some of the

subregions from this source are recognized and these, along with a number of new subdivisions, are designated as "sectors".

HADDOW, J. F., R. TRAUB & M. ROTHSCILD 1983

Distribution of ceratophyllid fleas and notes on their hosts. Material in the collection, with additional notes and maps of the genera. pp. 42-163 + 151 maps

This section treats geographic and host-distribution, arranged alphabetically by genus and species. Each generic account is accompanied by a small, unnumbered map showing the general distribution of the genus. Species accounts include the following information: name; authority; date; generic, specific and subspecific abbreviation (in cases where the species is the generitype this is indicated); map reference number; number of specimens mounted and if alcoholic material is present in the collection. Under this are listed the major host(s), general geographic distribution, ecological associations and host records. Under the latter heading, hosts are indicated as major(M), specific(S), occasional(O), or accidental(A). Also in this category, distinction is made between those host records based on material in the British Museum (N.C. Rothschild) collection and those drawn from the literature.

The 151 maps at the end of this section are printed in light blue. Many of the maps show the ranges of various hosts with collection records of their fleas superimposed in black symbols of different shapes. Where the ranges of more than 2 host species overlap, the combination of blue ink and reduction causes the maps to be extremely difficult to interpret.

TRAUB, R. 1983

The hosts of ceratophyllid fleas. pp. 164-187 with 9 tables in text.

This section is an exhaustive treatment of the origins, distribution and host preferences of both the Ceratophyllidae and the Leptopsyllidae. Both are mainly limited to the northern hemisphere and this treatment presents impressive evidence supporting the thesis that ceratophyllids evolved in the Nearctic Region as parasites of squirrels and their relatives, while leptopsyllids evolved in the Palaearctic Region and now mainly parasitize cricetine and microtine rodents.

TRAUB, R. & M. ROTHSCILD 1983

Evolution of the Ceratophyllidae. pp. 188-201

Many of the points which were made in the preceding section are reemphasized in this section with particular attention being directed to ceratophyllid morphology, taxonomy and phylogeny, physiology and behavior, and geological age. The authors conclude that the family is of Nearctic origin and originated in the early Oligocene (38-40 million years ago), co-evolved with squirrels, and only later did some become parasites of cricetid rodents and birds.

TRAUB, R. 1983

Medical importance of the Ceratophyllidae. pp. 202-228 with 2 tables in text.

After a brief introduction, the author sets down 23 factors affecting epidemiology as concerning outbreaks of epidemics or epizootics. These are followed by a consideration of the effects of these epidemiological factors. The various types of infectious agents (i.e. bacterial, rickettsial and viral) are then reviewed and discussed. It is concluded that ceratophyllid fleas are of little importance in the transmission of disease to humans, but that they play their most important role in maintaining cycles of infection in the reservoir hosts, particularly rodents.

The volume ends with 5 appendices. Presumably all 3 authors participated in these compilations, except for number 4 which was written solely by M. Rothschild. Their titles are self-explanatory.

1. Species described since 1975. p. 229
 2. Classification of hosts treated in text. pp. 230-232
 3. Alphabetical lists of hosts cited, showing their major ceratophyllid fleas (if any). pp. 233-252
 4. Distribution of British Ceratophyllidae. p. 253, plus maps 152-175
 5. List of taxa of Ceratophyllidae treated in this volume. pp. 254-268
- Index to fleas cited. pp. 269-288

Workers accustomed to the standardized format of the Catalogue volumes may find this volume difficult to use until they become familiar with its organization. However, in addition to the obvious usefulness of the generic key and the species accounts, there is a wealth of information contained in the other sections unparalleled in any other volume dealing with the order. The authors should be congratulated for their persistence and dedication to the production of this unique study.

---R.E.L.

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RECENTLY PUBLISHED - Entomology, a guide to information sources by Pamela Gilbert and Chris J. Hamilton, 1983. (ISBN 0-7201-1680-5) Mansell Publishing Limited, 6 All Saints Street, London N1 9RL, U.K. 237 pp. (Distributed in the United States and Canada by The D.H. Wilson Company, 950 University Avenue, Bronx, N.Y. 10452) US \$29.00

A frequent frustration among specialists is the difficulty of locating peripheral information relating to one's specialty. Entomology, being the broad discipline that it is, is particularly productive of such frustration. At last, in one relatively small volume, is a wealth of information on such diverse subjects as insects in art, literature and gastronomy; nomenclators and catalogues; dictionaries and glossaries; laboratory culture and mass rearing; sources of illustrative materials; books and review articles; computerized information retrieval; newsletters and conference calendars (Ms Gilbert is on the FN mailing list); learned societies and professional organizations; translation services and guides, and many other topics. As stated in the Preface "....the work is intended to be an introduction and source book for entomology by subject...", and this goal has been admirably attained. The contents are arranged in 8 chapters: 1. Introduction; 2. Naming and identification of insects; 3. Specimens and collections; 4. The literature of entomology; 5. Searching and locating the literature; 6. Keeping up with current events; 7. Entomologists and their organizations; and 8. Miscellaneous services. Each chapter or subdivision contains a brief introductory statement followed by consecutively numbered listings including annotations. Addresses are included for all organizations, a practice seldom followed in similar publications of this type.

The authors are to be congratulated for their devotion to the field of Entomology and their demonstrated good taste in including an illustration of a flea (Spilopsyllus cuniculi) on the cover.

---R.E.L.

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NEW JOURNAL - BIOLOGY 83: Current Titles in the Biological Sciences is a new quarterly periodical designed for the research-oriented professional in the biological sciences who wishes to keep abreast of the current literature. Biology 83 includes only titles published in 1983. Biology 84 will be limited to titles published in 1984. Contents are arranged first by subject matter such as Agriculture, Aquaculture, Behavior, etc. Authors are listed alphabetically under these headings as in the following example:

Torchio, P. F. 83-10530
SUBJ.: Biochemistry
TAXA: Hymenoptera: Megachilidae

A taxonomic index follows in which the example listed here appears as follows:

Hymenoptera: Megachilidae
Torchio, P. F. 83-10530
SUBJ.: Agriculture: Biochemistry

This is followed by a list of titles arranged alphabetically by author. In it the example appears as follows:

Torchio, P.F. 1983. The effects of field applications of naled and trichlorfon on the alfalfa leafcutting bee Megachild [sic!] rotundata (Fabricius) Jour. Kansas Entomol. Soc., 56(1): 62-68. Bee Biology & Systematics Laboratory, USDA-ARS, Utah State University, UMC 53, Logan, UT 84322. TAXA: Hymenoptera: Megachilidae
SUBJ.: Agriculture, Biochemistry 83-10530

Annual subscription rates in U.S. dollars for 4 issues per year are:

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Institutions	80.00	95.00	95.00

Additional information is available from S. R. Edwards, Editor, Biology 83, P.O. Box 368, Lawrence, Kansas 66044, U.S.A.

* * *

CONGRATULATIONS--Shri K. G. Narayana Pillai has been awarded the Ph.D. by the University of Kerala for a dissertation entitled "The post-embryonic development of the ovary of [the] rat fleas Xenopsylla cheopis and X. astia (Siphonaptera: Pulicidae).

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NOTES

YOU WILL NOTE that with this issue we have begun consecutive page numbering. This will apply only from FN 21, and has been instituted with an eye toward periodic indexing. Following is the page sequence which will be followed in future references to earlier issues.

Flea News 21	(December, 1980)	pp. 1-14
Flea News 22	(June, 1981)	pp. 15-28
Flea News 23	(November, 1981)	pp. 29-44
Flea News 24	(May, 1982)	pp. 45-60
Flea News 25	(November, 1982)	pp. 61-78
Flea News 26	(May, 1983)	pp. 79-98
Flea News 27	(December, 1983)	pp. 99-120
Flea News 28	(June, 1984)	pp. 121-140

NOTICE - As of 1-June-1984, Mr. Anthony Hutson will no longer be affiliated with the Siphonaptera Section of the British Museum (Natural History). We have not been informed of Mr. Hutson's replacement. Presumably, requests should be sent to Dr. Lawrence A. Mound, Keeper of Entomology.

ROBERT TRAUB has requested and received early retirement from the University of Maryland School of Medicine so that he can spend full time on research. Although he has been appointed a Research Professor at the same institution, it would be best if mail to him were henceforth addressed as follows:

Robert Traub, Ph.D.
Siphonaptera Section - NHB165
Dept. of Entomology
National Museum of Natural History
Smithsonian Institution
Washington, D.C. 20560
Phone: (202 357-1867

ROBERT S. GEORGE, who recently retired from teaching, has established a small entomological supply establishment specializing in nets, cages and Malaise traps. Interested parties may obtain a catalogue by writing Bob at the following address:

MARRIS HOUSE NETS
8 Saint Peter's Street
Duxford, Cambridge,
ENGLAND CB2 4RP
Phone: 0223-834979

DR. ROBERT PILGRIM, University of Canterbury, Christchurch, New Zealand, is seeking a copy of:

Smirnova, G. 1950

Sistematika lichinok nekotrich vidov bloch.
(Systematics of some species of larval fleas.)

Unpublished thesis. Entomology Department, Moscow University.

Anyone knowing of the availability of this work in the original or in translation may contact Dr. Pilgrim in the Department of Zoology.

* * *

CONFERENCE PAPERS

A symposium on Urban Entomology sponsored by the Society of Vector Entomologists will be held on June 13, 1984, in conjunction with the annual meeting of the Pacific Division, American Association for Advancement of Science, San Francisco, California, June 10-15. One paper on fleas is scheduled for presentation:

Fleas and their interaction with plague in the urban environment. - B. C. Nelson (for address, Flea News 21, p. 7)

Presented at the North Central Branch meetings of the Entomological Society of America, Wichita, Kansas, March 26-29, 1984:

Efficacy of twelve insecticides against the cat flea, *Ctenocephalides felis* (Bouché). - K. Schwinghammer, E. Ballard & F. Knapp, University of Kentucky, Lexington, Kentucky 40546.

Scheduled for presentation at the annual meeting of the American Society of Mammalogists, June 24-28, 1984, Humboldt State University, Arcata, California:

The relationship of habitat use and flea infestation in meadow voles (*Microtus pennsylvanicus*) and white-footed mice (*Peromyscus leucopus*). - W. C. Berseth & D. M. Cameron, Dept. of Biology, York University, Downsview, Ontario, Canada M3J 1P3.

* * *

MAILING LIST

ADDITIONS

- A. J. E. Harman, 20 Chestnut Close, Hockley, Essex SS5 5EJ, England
B. K. Kotti, Parasitological Laboratory, All-Union Anti plague Institute "Mikrob", Sovietskaya 13, Stavropol 355106, U.S.S.R.
Ms. Algenir Ferraz Suano de Silva, Chefe da Divisão de Biblioteca, Instituto Nacional de Pesquisas ca Amazonia, Manaus, Amazonia, Brazil

CHANGES

- Prof. G. M. Dunnet, University of Aberdeen, Zoology Department, Culterty Field Station, Newburgh, Ellon, Aberdeenshire AB4 0AA, Scotland
Dr. N. D. Emel'yanova, Donskaya Street 21, Flat 6, Irkutsk 31, U.S.S.R.
Mr. R. S. George, Maris House, 8 St. Peter's Street, Duxford, Cambridge CB2 4RP, England
Dr. J. M. Klein, Centre Pasteur, B. P. 1274, Yaoundé, Cameroun, Afrique
Mr. S. M. Kulkarni, National Institute of Virology, 20-A Dr. Ambedkar Road, Post Box No. 11, Pune - 411 001, Maharashtra State, India
Mr. R. Mehl, Laboratory of Medical Entomology, National Institute of Public Health, Postuttak, N-Oslo 1, Norway
Dr. Y. Schlein, The Hebrew University, Dept. of Parasitology, Hadassah Medical School, P.O. Box 1172, Jerusalem, Israel
Dr. R. C. H. Shepherd, Keith Turnbull Research Institute, Box 48, Frankston, Victoria 3199, Australia

DELETIONS

- | | | |
|---------------------|-----------------------|-------------------|
| Dr. M. H. Abreu | Dr. M. J. Cotton | Dr. E. V. Isayeva |
| Dr. V. Aellen | Dr. N. A. Gershkovich | Dr. R. Iyengar |
| Dr. A. G. Avetisyan | Dr. G. P. Holland | Dr. S. Kristensen |
| Dr. S. Brelih | Dr. Harry Hoogstraal | Dr. D. G. Krylov |
| Dr. J. Cooreman | Mr. A. M. Hutson | Dr. J. Lachmajer |

(1973 - 1976)

DELETIONS from Mailing List - continued

Dr. Michael Liantas	Dr. K. Rai	Dr. V. E. Tiflov
Dr. C. E. Machado-Allison	Prof. Dr. B. Rosický	
Dr. H. Naing	Mr. Rick Sangerman	Mr. J. A. Vaughan
Dr. Z. Ono	Mr. K. N. Singh	Dr. N. A. Violovich
Miss A. M. Pearce	Mohamed Sofi	Dr. J. Viramo
Claude Pratt	Dr. E. F. Sosnina	Dr. S. A. Vysotakaya
Dr. V. N. Prokop'yev	Dr. A. N. Talybov	Dr. V. I. Yurkina
		Prof. Dr. I. F. Zhovtyi

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PROVISIONAL LISTS OF TAXA DESCRIBED IN 1982 AND 1983---

There are no additions to these lists at this time. Readers are urged to inform us of any omissions. Our library has not yet received all issues for 1982 for some journals. Consequently, numbers will not be assigned to 1982 taxa until Flea News 29.

* * *

An asterisk (*) preceding a citation indicates we have seen neither the original publication nor verified it in another source. Anything appearing in brackets ([]) after a citation indicates an annotation, e.g., the name of a taxon described.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1973 (list 14)

*Ahn, Y. K. & C. T. Soh - Flea fauna of rodents in coastal region of Korea. Yonsei Rep. trop. Med. 4(1): 41-49

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1974 (list 17)

Anderson, J. R. - Symposium on reproduction of arthropods of medical and veterinary importance II. Meeting of the sexes. J. med. Ent. 11(1): 7-19 [Siph.: 9-12]

Frazier, C. A. - Biting insect survey: a statistical report. Ann. Allergy 32: 200-204

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1975 (list 17)

Mesina, J. E. & R. S. F. Campbell - Wild rodents in the transmission of disease to animals and man. Vet. Bull. 45(2): 87-96

*Thornton, D. J., R. C. Tustin, B. J. Pienaar et al. - Cat bite transmission of Yersinia pestis infection to man. J. S. Afr. Vet. Assoc. 46: 165-169

*Turner, R. W., P. Soeharto & M. Supalim - Dynamics of the plague transmission cycle in central Java: ecology of mammalian hosts with special reference to Rattus exulans. Bull. Hlth. Stud., Indonesia 3: 41-71

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1976 (list 16)

Arthur, D. R. - Interaction between arthropod ectoparasites and warm blooded hosts. In: Kennedy, C. R., ed., Ecological aspects of parasitology. North Holland Publ. Co., Amsterdam, pp. 163-183

(1976 - 1979)

*Kadulski, S. & J. Dobrynczuk - Observations on the ectoparasites of the European hare. In: Ecology and management of European hare populations. Warsaw

Topachevskii, V. A. - Fauna of the USSR: Mammals. Mole rats, Spalacidae. Vol. 3, no. 3. Acad. Sci. USSR, Zool. Inst., n.s. 99, Leningrad, 308 pp. [Siph.: 97] [English transl. publ. Amerind Publ. Co. Pvt. Ltd., New Delhi]

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1977 (list 14)

*Bavasan, A., S. Covelma & D. Eregdendagva - Fauna bloch Mongolskoj narodnoj respubliky. Erdem Sinžilgeezaach Argen bičig. No. 1[XII]: 107-123

Friend, W. G. & J. J. B. Smith - Factors affecting feeding by blood-sucking insects. Ann. Rev. Entomol. 22: 309-331 [Siph.: 325-326]

Nelzina, E. N. - Main taxonomic groupings of organisms participating in the formation of nest-burrow microbiocenoses. Parazitologiya 11: 326-332

Newson, H. D. - Arthropod problems in recreation areas. Ann. Rev. Entomol. 22: 333-353 [Sylvatic plague: 346-347]

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It will be understood that- although their titles are given in English in these lists - papers in Russian or Chinese periodicals, or books published in the USSR or China, are in Russian or Chinese (but often with summaries in English).

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